

Crop Production III / CPT 431

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1. COCOA

Kritica

Cacao, (*Theobroma cacao*), also called cocoa, tropical evergreen tree (family Malvaceae) grown for its edible seeds, whose scientific name means "food of the gods" in Greek. Native to lowland rainforests of the Amazon and Orinoco river basins, cacao is grown commercially in the New World tropics as well as western Africa and tropical Asia. Its seeds, called cocoa beans, are processed into cocoa powder, cocoa butter, and chocolate.

Cocoa is an important commercial plantation crop of the world. Cocoa is a crop of humid tropics and so it was introduced as a mixed crop in India in areas where the environments suit the crop. It is cultivated in coconut and arecanut plantations large scale from 1970 onwards.

Climate and soil

The natural habitat of the cocoa tree is in the lower storey of the evergreen rainforest, and climatic factors, particularly temperature and rainfall, are important in encouraging optimum growth. Cocoa is a perennial crop, and it can withstand different seasonal variations with good health and yield potential. Cocoa is normally cultivated at altitudes up to 1200 m above MSL with an annual rainfall of 1000mm to 2000mm and a relative humidity of 80 % with maximum 350C and minimum temperature of 150C. Cocoa can be grown as intercrop in coconut and arecanut gardens. It is predominantly grown on red laterite soils. It thrives well on wide range of soil types with pH ranging from 4.5-8.0 with optimum being 6.5-7.0.

Varieties

There are three varietal types in cocoa namely Criollo, Forastero and Trinitario. Forastero types are known to perform well under Indian conditions. Kerala Agricultural University has released 7 improved clones of Forastero types namely CCRP - 1, CCRP - 2, CCRP - 3, CCRP - 4, CCRP - 5, CCRP - 6 and CCRP - 7 and 3 hybrids CCRP - 8, CCRP - 9, CCRP - 10. Central Plantation Crops Research Institute.

Propagation

Cocoa can be propagated by seeds. Seeds are to be extracted from pods. Cocoa pods take 150-170 days from pollination to attain the harvest stage. The stage of maturity is visible from the change of pod colour from green to yellow (Forastero) and red to yellow (Criollo).

Criteria for selection of mother plant for collection of seeds

1. Forstere type (green-immature, yellow-ripe) having medium to large pods of not less than 350 g weight, smooth or shallow furrows on the surface without prominent constriction at the neck can be selected.
2. Husk thickness of pods should be less than 1 cm.
3. Pod value (number of pods to give 1 kg wet beans) should be less than 12.
4. Number of beans per pod has to be more than 35.
5. Bean dry weight to be more than 1 g.
6. The best seeds for sowing are those from the middle of the pod.
7. Harvested seed pods can be stored in shade up to one week. Viability of the beans can be extended for some more days if freshly extracted seeds are stored in moist charcoal and packed in polybags. Other alternative is extracting beans, removing the testa and packing in polythene bags.

Potting media

Red Soil + Sand + FYM @ 2:1:1 + Super phosphate (5 kg/ton) was found to be best media for growth and development of cocoa seedlings

Vegetative propagation

As the seedling progenies showed wider genetic variability, to maintain true to types, asexual or vegetative propagation is followed. Grafting and budding are being followed in multiplication of cocoa. It also ensures multiplication of identified high yielding clones in large quantities. Though vegetative propagation of cocoa by budding, rooting of cutting and grafting are feasible, the widely accepted methods in India are budding and grafting.

Selection of planting materials

(Selection of Planting materials)

When seedlings are used as planting, select vigorous and healthy seedlings from polyclonal garden. The planting material should be of 4-6 month old seedling or grafted or budded plant. The seedling/grafted/budded plant should be planted in the centre of the pit, not too deep. While planting grafts, polythene strip tied over graft joint should be removed and the joint should be above the soil.

Time and method of planting

Cocoa is a shade loving plant. During its seedling period it requires about 50% shade and later the shade requirement is about 40%. The plant is grown as a mixed crop with other plants like spices and rubber mainly under rainfed conditions. Cocoa is planted as an intercrop in coconut and arecanut gardens.

One Cocoa row of plants at 3m interval in the center of two Coconut rows and One Cocoa at the center of two Coconut plants along the Coconut row. The layout will hold about 500 plants per

Prepare PHS of 50 cm x 50 cm x 50 cm or less above the weather for one month are reflected with topsoil and 15-20 kg of compost of FYM to ground level. Tear off the polybags carefully. Place the soil ball with the seedlings in the growing holes with minimum disturbance and press the soil around firmly.

Irrigation in cocoa

Cocoa is usually grown in areas where water availability is adequate. Cocoa plants are sensitive to drought, irrigation in such cases becomes essential. During summer, as it exists in Southern India, the crop requires irrigation at weekly intervals. When it is grown as mixed crop with arecanut, the crop is to be irrigated once in a week during November-December, once in 6 days during January-March and once in 4-5 days during April-May with 175 litres of water.

Age of the plant	Water requirement (litre/ plant / day)
1st year	3-5
2nd year	10
3rd year and later	20-25

Soil nutrient management

Soil nutrient management is critical to the general health of the tree, particularly where cocoa is grown on poor soils with low nutrient levels. The fertility of soils under cocoa plantations with complete canopy formation can be maintained or sustained for a fairly long time due to the ability of cocoa fallen leaves to recycle nutrients back into the soil and decomposition of leaf litter. However, continuous harvesting will eventually result in loss of soil nutrients

Fertilizer schedule for cocoa

An annual application of the following schedule should be applied in two equal splits, the first dose in April- May and the second dose in September- October i.e. pre and post monsoon applications.

Fertilizer (g/ plant)	I year	II year	III year onwards
Urea	72	144	220
Rock phosphate	65	130	200
Muriate of Potash	77	154	230

Method of application Fertilizer may be applied uniformly around the base of the tree up to a radius of 30 cm during the first year, forked and incorporated into the soil. For grown up plants the best method is to rake and mix the fertilizers with soil in shallow basins of around 75 cm.

These ratios may be increased gradually up to 1:1.5 one year after first year. Care should be taken not to spill the inorganic fertilizers on the trunk, branches or leaves of young trees in order to avoid burning.

Nutrient management in cocoa

Sixteen plant nutrients are essential for proper crop development. Each is equally important to the plant, yet each is required in vastly different amounts. These differences have led to the grouping of these essential elements into three categories: primary (macro) nutrients, secondary nutrients, and micronutrients.

Primary (macro) nutrients

Primary (macro) nutrients are nitrogen, phosphorus, and potassium. They are the most frequently required in a crop fertilization program. Also, they are used in the greatest total quantity by plants as fertilizer.

Secondary Nutrients

The secondary nutrients are calcium, magnesium, and sulphur. For most crops, these three are needed in lesser amounts than the primary nutrients. They are growing in importance in crop fertilization programs due to more stringent clean air standards and efforts to improve the environment.

Micronutrients

The micronutrients are boron, chlorine, copper, iron, manganese, molybdenum, and zinc. These plant food elements are used in very small amounts, but they are just as important to plant development and profitable crop production as the major nutrients. Especially, they work "behind the scene" as activators of many plant functions like chlorophyll production, carbohydrate formation, cell division and starch formation etc.

Micronutrient deficiency

Zinc Deficiency

Symptoms are chlorosis of the leaves. This appears in patches and in advanced stages the green areas are found only along the vein margins, giving a 'vein banding' appearance to the leaves. Affected leaves show mottling and crinkling with wavy margin. Most of the younger leaves become narrow and sickle shaped showing characteristic 'little leaf' symptom.

Symptoms on twigs include rosette and dieback. Shortening of internodes causes a rosette type of growth.

Management: Foliar spray of a mixture of 0.36% (3 gm in 10 liter of water) Zinc Sulphate and 0.15% (w/v) lime.

Training and pruning

Pruning and shade management are ~~essential~~ ^{essential} in cocoa farming involves thinning of branches and removal of old or dead stems, whilst shade management involves leaving forest trees and/or planting shade trees to ~~optimize~~ ^{optimize} the light intensity in the cocoa grove. Pruning serves many purposes, including:

1. It determines the shape of the tree. ~~This important that the tree is shaped to~~ facilitate local

Management practices:

1. It maximizes the nutrient distribution towards pods. By cutting away new and unproductive chupons on mature trees, the pod size will be increased.
2. Thinning the cocoa canopy causes more light to filter to the centre of the tree, and more air circulation, whereby Black Pod disease can be reduced.

Pruning and height control

The basic aim of pruning cocoa trees is to encourage a tree structure that allows sunlight to filter through to the main branches and trunk (what is known as a jorquette) to stimulate flowering and facilitate harvesting. Pruning generally takes place twice a year. The first pruning is done after the main harvest (from April to July) and just before the rainy season, and the second pruning five months later, during the months of November and December. Young plants should develop a jorquette at a height of about 1 meter. It has been found that increasing light intensity decreases the jorquette-height. If a jorquette is considered too low, it can be cut off. The strongest of the re-growing chupon can be selected and all others removed. In due course, this chupon will produce a jorquette at a higher level. All prunings should be left in the field to rot down, except the diseased ones. Diseased or unwanted branches with vascular die-back and water shoots are to be removed to maintain the health and vigour of the trees.

Pruning of grafted plants

Pruning of grafted plants is done, after first year of planting, primary pruning should be done to obtain a supporting framework of one or more upward growing main stems. Then drooping or inward growing branches are to be removed. Secondary pruning is suggested to develop well-shaped canopy and desired canopy should be maintained in umbrella shaped form with about 3.8 m to 4.2 m spread and 2.7 m height depending upon the space and main crop in which cocoa is under planted/grown. Pruning is usually done annually in August-September.

Plant protection in cocoa

Pests and diseases are important risk to productivity and quality of harvest which in turn affects the returns to the farmers. Since cocoa is an introduced crop the more important for the farmer is to be clear about the pests and diseases and to take it to identify the symptoms correctly.

Diseases

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1. Seedling blight (*Phytophthora palmivora*)

The symptom develop on the leaves and stem of seedlings or budded plants. On leaves, small water soaked lesions appear which later coalesce in the blighting of leaves. On stem, water soaked lesions develop initially and later turn to black colour. Stem infection develop any point on the stem causing the death of seedlings. Management: Remove and destroy the affected seedlings. Spray with 1 percent Bordeaux mixture or 0.2% copper oxychloride just before the onset of monsoon and thereafter at frequent intervals.

2. Black Pod rot (*Phytophthora palmivora*)

Infection appears as chocolate brown spot, which spreads rapidly and soon occupies the entire surface of the pod. As the disease advances, a whitish growth of fungus consisting of fungal sporangia is produced over the affected pod surface. Ultimately, the affected pods turn brown to black. The internal tissues as well as the beans become discolored as a result of infection. The beans in the infected pods approaching ripeness may escape infection because they are separated from the husk on ripening.

~~Management:~~ Periodically remove the infected pods. Spray 1 percent Bordeaux mixture with on set of monsoon and also frequent intervals. Provide frequent drainage and regulate shade to increases aeration. *Pseudomonas fluorescens* (PT1) liquid formulations @ 0.5% as soil and foliar spray (3 times per year- June, October & February) was found to be effective in reducing the Cocoa Pod rot and Stem canker.

Preparation of 1% Bordeaux mixture

The mixture of copper sulphate and lime was known as Bordeaux mixture

Procedure:

1. Dissolve one kilogram of copper sulphate in fifty liters of water in a plastic bucket
2. Dissolve one kilogram of lime in fifty liters of water separately in a plastic bucket
3. Pour copper sulphate solution into the lime water slowly with constant stirring using a wooden stick
4. Test the mixture before use for the presence of free copper, which is harmful to the plant by dipping a polished knife in it.

Harvesting and Processing

Cocoa produces flowers from the 3rd year of planting onwards and economic yield starts from 5th year. Properly well maintained (irrigation/nutrient management) tree yielded 1-2 kg of pods / tree / year. Pods take about 140 - 160 days to ripen. Generally, cocoa gives two main crops in a year, i.e. September - January and June to August. Offseason crops may be seen throughout the year especially under irrigated condition. The stage of maturity of the pod is best judged by change of color of pods. Pods that are green when

Immature turn yellow when mature and redish pods turn yellow or orange. The change in color starts from the grooves on the pods and then spreads to the entire surface. The harvesting is to be done at regular intervals of 10-15 days. Avoid over ripening of pods.

The pods are heaped together and kept under shade for 5-6 days to ensure uniform ripening. This reduces acidity and helps in development of a better flavor suited for chocolate making. Cocoa pods are collected at a central location, where pods are broken, husks removed and the white-yellowish seed masses are heaped together for fermentation. Fermentation takes about 5 to 7 days, depending on the season and temperature. Farmers sometimes mix the heap on the 2nd or 3rd day, to allow for aeration and a more uniform fermentation. For breaking the pods, wooden billets may be used. After breaking the pods crosswise, the placenta should be removed together with husk and the beans are collected for fermentation. On an average 10-12 pods give 1 kg wet beans and 3 kg of wet beans (from 30-36 pods) give 1 kg of fermented and dried beans. Under normal cultivation practices, each cocoa tree yields about 1-2 kg annually.

Fermentation

Fermentation of cocoa beans is essential to remove the edhering mucilaginous pulp to develop flavour and aroma precursors, reduce bitterness and kill the germ of the seed and to loosen the testa. The process is simple but must be carried out properly in order to get beans for good quality. Different methods of fermentation normally followed are 1. Box 2. Basket. However, in this box and basket methods are recommended depending on quality of bean to be fermented.

Box method

The boxes of 60 cm X 60 cm X 45 cm made of wood and having reapers at the bottom to allow the sweating from the pulp to drain out and provide aeration are used. The boxes could be arranged in tiers for transferring beans from one to the next in line below. Two detachable wooden planks are provided on one side of the box for transferring (mixing) the beans by removing the planks. The beans are loaded on fermentation box and covered with banana leaves or gunny bags. The mixing of beans is effected while transferring to the next box after 24 hours. The temperature of the fermentation mass will rise to 45 - 55°C after about 48 hours of fermentation. Every alternate day the beans under fermentation have to be properly mixed for uniform fermentation. This has to be continued for 6 days.

Basket methods

In this method, Bamboo or cane baskets of suitable size could be used for fermenting small quantity of beans. One or two layers of banana leaves are placed at the bottom with provision to drain the sweating. The basket is filled with the beans and the surface is covered with banana leaves. A small weight is placed over the banana leaves. The basket is placed over a raised surface to facilitate drainage of the sweating for one day. Later the basket is covered with thick gunny bags. The beans are mixed thoroughly on the 3rd and 5th days and again covered with gunny bags. The fermentation will be completed at the end of 6th day and the beans withdrawn for drying.

Drying

During fermentation the cocoa flavour develops and the beans turn brown. After fermentation, the beans can be dried by sun drying. The fermented cocoa beans have considerable moisture (55-65 %) and the drying rate is depend upon temperature and the airflow. Sun drying should be adopted as far as possible, as it gives superior quality produce compared to that by artificial drying. The moisture content of well dried beans is around 6-7 %. Slow drying is preferable for better quality of the beans.

Grading and storage

The flat, slate, shrivelled, broken and other extraneous materials are removed. The cleaned beans are packed in fresh polythene lined (150 - 200 gauge) gunny bags. The bags are kept on raised platform of wooden planks. The beans should not be stored in room where spices, pesticides and fertilizers are stored as they may absorb the odour from these materials.

COFFEE

The coffee plant

Botanically coffee belongs to the family Rubiaceae along with gardenias and guinine. The coffee plant can grow to a great height and because of that it is described as a coffee tree. In the wild it can grow as high as 10 metres. On plantations and coffee producing farms, they are pruned to a height which is most comfortable for harvesting.

The flowers of the plant are white and smell like jasmine and orange. Coffee plants flower at different times: Coffea arabica flowers after rain and Robusta flowers irregularly.

The coffee fruit is the size of a cherry and when ripe the colour is usually dark red or yellow. The berries are ripe for picking about nine months after flowering. They don't all ripen at the same time, so a single tree has to be picked several times. Each fruit contain two coffee beans. Coffee beans grow pressed together inside the berries.

There are more than 60 varieties of coffee but only two of them have commercial importance:

- Coffea Arabica
- Coffea Canephora – Robusta

Arabica is originally from Ethiopia and today it is grown around the world. Robusta was discovered in Congo and from there it has spread throughout tropical Africa and to Indonesia, Vietnam and beyond. About 65 % of the world's output is Arabica, and 35 % is Robusta. Arabica's flavour is milder and has more nuances than Robusta. The strong flavour of Robusta makes it best suited to blends of dark roasted coffees like espresso or French-style coffees.

Coffee growing areas and countries

Coffee grows in the tropics, the region between the Tropics of Cancer and Capricorn, in different parts of the globe. There are five major areas where coffee is cultivated:

- Central America
- Africa
- Asia
- Oceania

Coffee production. Cultivation of the coffee plant, usually done in large commercial operations. The plant, a tropical evergreen shrub or small tree of African origin (genus *Coffea*, family Rubiaceae), is grown for its seeds, or beans, which are roasted, ground, and sold for brewing coffee. This section treats the cultivation of the coffee plant.

The Arabica species of coffee is cultivated mostly in La in America, while the Robusta species predominates in Africa. Both coffee species are grown in India, Indonesia, and other Asian countries. There are many varieties, forms, and types of each. The effects of environment and cultivation further increase this diversity.

Climatic factors most important for coffee growth are temperature and rainfall. No variety can withstand a temperature in the vicinity of 32°F (0°C). Temperatures between 73° and 82° F (23° and 28° C) are the most favourable. Rainfall of 60 to 80 inches (1,500 to 2,000 millimetres) per year is required along with a dry period of two to three months for the Arabica. Irrigation is required where annual rainfall is less than 40 inches (1,000 millimetres).

Plantations are usually established in cleared forestland. The young shrubs are planted in rows spaced so that the density varies between 500 and 750 plants per acre (1,200 and 1,800 plants per hectare). Seedlings or cuttings raised in nurseries are carefully planted at the beginning of the rainy season; until they start producing fruit three to four years later, their care is limited largely to the trimming required to give them a robust, balanced framework and to stimulate fruiting.

Seeds and sowing

Coffee is propagated by seeds

Season

Planting spreads from June - December

Preparation of seeds

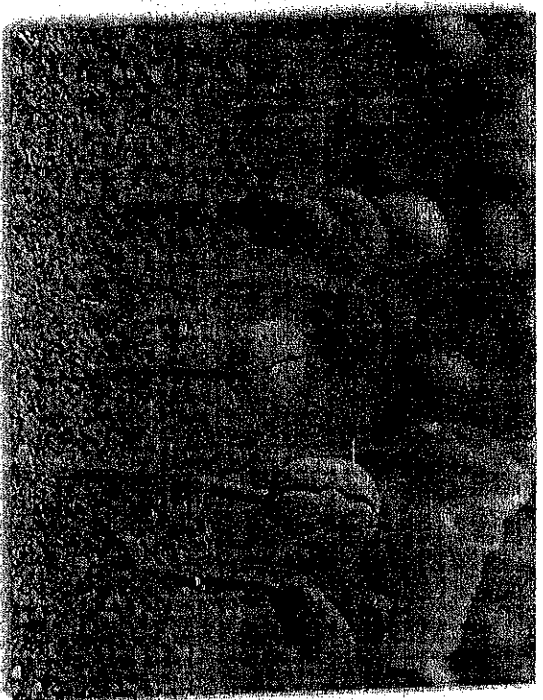
Healthy and well developed fully ripe berries are harvested from specially identified plants for use as seed bearers. After discarding the floats, the sound fruits are de pulped, sieved and mixed with sieved wood ash and dried in shade. The seed is then graded to remove all cut, triangular and elephant beans. Prior to planting, the seeds are treated with Agrosan or any Organomercurial compound to prevent fungal infection.

Nursery practices

Select light loamy soil of good drainage with high organic matter content with water and shade facilities. Form raised beds of 15 cm height, 1m width and at convenient length. Incorporate 30 - 40 kg of well rotten compost, 2 kg of finely sieved agricultural lime and 400 g of rock phosphate to a bed of 1 x 6 m size. In heavy soils, it is necessary to add coarse sand for drainage and aeration.

Sowing

Pre-sowing seed treatment with *Ascopryl* and *Phosynobacterium* can be done. Seeds are sown in December - January in the bed 1.0 - 2.5 m apart with the flat side down, weeds in regular rows. Then they are covered with a thin layer of fine soil and a layer of paddy straw. Water the beds daily and protect from direct sunlight by an over head jandaal. Seeds germinate in about 45 days after which they are transplanted to a secondary nursery beds for raising ball or Bag nursery.



Bag nursery

Polythene bags with adequate number of holes in the bottom half are taken and are filled with a prepared mixture containing jungle soil, FYM and sand in the proportion of 6:2:1. An area of 12 x 8 m can accommodate 5000 seedlings. Seedlings are planted in polythene bags.

Preparation of field

Selective felling may be done while retaining a number of desirable shade trees. Terracing should be done in deep slopy areas. After the summer showers, pits of 45 cm x 45 cm x 45 cm are dug at 1.25 - 2.5 m apart. The pits are left open for withering and then filled and heaped for planting. At the time of filling, apply 500 g of rock phos mate per pit along with top soil. Planting is done along the contour in slopy areas.

Spain

Arabica Coffee 1.5 to 2.0 l either 25%

Dwarf varieties Santaracha 1 x 1 m.

Robusta coffee 1 2.5 m either way.

Planting shade trees

Dadap is commonly used as a lower canopy shade. Two metre long stakes are planted for every two plants of coffee. Silver Oak and Dadaps are planted during June when rains of South-West monsoon commences.

Irrigation

It is generally grown as a rainfed crop. But irrigation with sprinkler during March - April increases blossoming and results in higher yields.

Diseases

Among the diseases of the coffee shrub are leaf rust caused by the fungus *Hemileia vastatrix*, which does considerable damage in the plantations of Arabica and the coffee berry disease caused by the fungus *Colletotrichum coffeum*, which also attacks the Arabica. Robusta appears to be resistant, or only slightly susceptible, to these scourges. Among the numerous parasites that attack the coffee shrub is the berry borer (*Stephanoderes hampei*), which damages the seeds of both Arabica and Robusta.

Rust

Spray 0.5% Bordeaux mixture in February - March (Pre-bloom) followed by 0.03% Oxycarboxin in May - June (Pre-monsoon). Repeat in July - August (mid-monsoon) September - October (Post-monsoon) with any one of the above fungicides or Spray 0.5 % Bordeaux mixture during the month of June followed by 0.02 % Triadione for curing September and 0.5 % Bordeaux mixture during the month of December.

Black rot or Koleroga

Centering and handling of the bushes should be done prior to the onset of South-West monsoon. Remove affected twigs. Spray 1% of Bordeaux mixture curing break in monsoon.

Cellar rot

Treat seeds with Carbenrazim 1 g/kg or Carboxin 0.7 g/lg. Maintain filtered shade in nursery. Drench nursery beds with Mancozeb or Captan 0.5 g/lit before sowing.

Brown eye spot

Brown eye spot can be controlled by spraying Captan or Mancozeb or Ferbam 2 g/lit or Carbenrazim 0.5 g/lit during September.

~~Black root rot~~

Dig out and burn infected bushes. Dig a trench 10 cm deep around affected spot along with a ring of healthy bushes. Prune the healthy bushes within and outside the trench to allow sunlight. Keep the trench free from fallen leaves. Do not replant for 18 months.

Harvest

Harvest starts during November and harvesting extends up to February. Coffee fruits should be harvested as and when they become ripe. Coffee is just ripe when on gently squeezing the fruits the beans inside come out easily. Unripe fruits should be scrupulously sorted out before using the fruits for pulping. They may be dried separately as cherry.

Fly picking: Small scale picking of ripe berries during October to February

Main picking: Well formed and ripened berries are harvested during December. Bunches of the yields are obtained from this picking.

Stripping: Picking of all the berries left irrespective of ripening.

Cleanings: This is collection of fruits that have been dropped during harvesting.

Unripe fruits should be scrupulously sorted out before using the fruits for pulping. They may be dried separately as cherry.

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3. OIL PALM

Botany. The oil palm tree (*Elaeis guineensis*) is a member of the family Palmae, subfamily Coccoideae (which also includes the coconut), genus *Elaeis*. The genus contains two main species: *E. guineensis* of African oil palm, and *E. americana* of American oil palm; the latter is only valuable for hybridization. Male and female inflorescences occur on the same tree in alternated cycles of the same sex, and are only differentiated after approximately two years. This process is influenced by moisture and temperature conditions, fertilization and other secondary ecological factors.

Oil palm (*Elaeis guineensis*) is a tropical tree crop which is mainly grown for its industrial production of vegetative oil. It is a typical estate crop, grown and harvested over large uniform areas (3,000 to 5,000 ha) around a central oil mill to allow rapid industrial handling after harvesting. Palm trees can also be observed in village gardens where they provide oil for local consumption, but in that case both yield and oil quality are much lower. Oil palm is a typical crop of the rainy tropical lowlands. The tree requires a deep soil, a relatively stable high temperature and continuous moisture throughout the year. Soil fertility is less important than physical soil properties. Dry periods of more than 2-3 months do not specifically damage vegetative growth, but affect seriously the production and quality of the fruit bunches.

Origin and Distribution. The origin of oil palm points to Africa, in particular to West Africa. Fossil pollen, similar to the oil palm as it grows today, have been found in Miocene and more recent strata in the Niger delta. Portuguese explorers of the Guinea coast mention the existence of trees appearing to be oil palms as early as 1434. In 1508 already reference has been made to palm groves in Liberia, and to palm oil trade near the Forcados River in Nigeria. Later, Portuguese, Dutch and English travelers refer to palm wine and palm oil in the area. The centre of origin of oil palm is the West and Central African coastal belt between Guinea and northern Angola. The palm spreads from 16° North in Senegal to 15° South in Angola, and eastwards to Zanzibar and Madagascar. The best production levels are attended in the high rainfall areas between 7° North and South from the Equator. Oil palms are sporadically encountered up to St. Louis in Senegal and to the Upper Niger valley near Bamako. Minor palm groves are observed around Dakar and in the Gambia, but their production is too low to justify the establishment of commercial oil mills. The real palm belt in Africa runs through the southern latitudes of Guinea, Sierra Leone, Liberia, Ivory Coast, Ghana, Togo, Benin, Nigeria, Cameroon and into the equatorial regions of Equatorial Guinea and the Congo. Cultivars and Classification. Cultivars in the strict sense do not occur. As the oil palm is monocious and cross-pollinated, individual palms are usually very heterozygous, and vegetatively propagated clonal material can not be made.

Palm

The current classification of cultivars is mainly based on fruit structure and seed (or commercial value):

• Macrocarpa:

shell (endocarp) 6-8mm thick; is an extreme form of Dura, which is still widely spread in Sierra Leone and western Nigeria; without any commercial value;

• Dura: shell

2-8mm thick, comprising 25-55% of weight of fruit; medium mesocarp content of 35-55% dry weight, but up to 65% in Deli palms; less productive but jerdy variety, well adapted to village gardens;

• Pisifera:

shell-less, with small pea-like kernels in fertile fruits; of little commercial value, because of its high abortion ratio, but important for cross-breeding commercial palms; and

• Tenera:

shell 0.5-3mm thick; comprising 1-32% of weight of fruit; medium to high mesocarp content of 60-95%, but occasionally as low as 55%; this variety is the result of a hybridization of Dura and Pisifera, and has a high commercial value.

~~Ecology and Growing Conditions~~. Oil palm is a typical tree crop of the tropical rainforest. It can however hardly survive or regenerate in dense secondary forest because of the lack of sunshine. This is also the reason why stand-alone trees in villages are generally much taller than in palm groves. For optimal growth and production the tree requires stable climatic conditions, in particular with respect to light and moisture supply. Any deviation from these conditions enhances a yield decrease. Oil palm thrives best in lowlands below 300-400m altitude.

Climatic Requirements - Temperature requirements for oil palm are simple and straight forward. High temperatures are favorable with as less as possible fluctuations. Below 18° C growth is stopped. The optimal daily temperature is 27-28° C, with mean monthly values between 30-32° C maximum and 21-24° C minimum. The mean minimum temperature of the coldest month should be > 18° C. Seeding growth is arrested at temperatures < 15° C.

Rainfall and moisture - Oil palms require an average rainfall of 150 mm/month, with dry periods not exceeding 2-3 months. Relative air humidity should be above 75% throughout the year. On average, a minimum annual rainfall of 1800mm is considered optimal, ranging up to 2,500mm without harm. Rainfall above 2,500mm is considered unfavorable because this interferes with a lower solar radiation. In Malaysia it is found that 2,000mm rain per annum produces the best yields. On Mount Cameroon oil palm production is possible in high rainfall areas (over 4,500 mm/year) because rains concentrate in short but heavy afternoon showers

leaving sufficient sunshine hours to benefit the oil palm. Oil palms can tolerate some flooding, provided the water is not stagnant and the water table is fluctuating.

Soil Requirements. Oil palm can be grown on a wide range of soils. Soil physical properties, and soil moisture in particular, are more important than nutrient supply, which can anyway be corrected by fertilizer application. The crop should be planted, wherever possible, on flat or undulating land: steep slopes increase both the risk of erosion and the cost of establishment and production, including the more difficult and more costly construction of access roads.

Soil physical characteristics — Oil palm requires deep and easily penetrable soils with a good moisture retention and free drainage. The primary root descends deeply from the base of the trunk, but remains short when the water table is high. Most finer secondary roots are in the top 1m of soil. The crop supports water logging only for short periods.

Planting and Land Management. Field Lay-Out and Plant Density — Field size corresponds usually with blocs of 100ha, separated by one main N-S oriented road and 4 intermediate E-W roads every 250m. The accepted norms for the road network are : 20m of 8-meter wide link roads (asphalted tracks) and 40m of 5-meter wide collection paths per hectare of plantation. The field lay-out should mainly be carried out in the dry season. It should also take into account that FFB bunches should not be carried more than 200m to the road where they are picked up for further transport. The plant population density is targeted at 143 trees/ha, but because of possible losses a minimum of 120 palms is needed. Optimal results are obtained by a triangular (9 by 9m, and 7.80m between the lines) or square (at 9m spacing) planting patterns. Yields diminish rapidly with higher density due to increased competition for light, water and nutrients.

Cover Crop - Once established in the field the space between the trees is rapidly occupied by a cover crop, either Calopogonium, Centrosema, Pueraria or Mucuna bracteata. All of these have the advantage that they rapidly establish and thus protect the surface, provide a high amount of organic material to the soil, avoid weed development, and serve as a nitrogen supplier to the crop.

Fertilizers- The exact amount of fertilizers to be supplied depends in the first place on the nutrient status of the soil before planting. Nitrogen dressings are important in the early years and are usually in the order of 0.22kg N per palm in the planting year, and 0.45kg per year for the following 3-4 years. Phosphate deficiency is rare. Potassium requirements are high, because large quantities of K are removed in the fruit bunches. Where old plantations become deficient in K, dressings up to 4.5kg KCl per tree over several years may be needed for full responses

diseases

The following is a list of major diseases encountered in oil palm nurseries in the Far East and Africa:

•freckle, caused by the fungus *Ceratocorys candelabrum*, creates orange-brown spots on the lower leaves, culminating in the drying out of leaf tips and edges. It occurs mainly in the middle and at the end of the dry season.

•blast disease, apparently associated to poor water management in the polybags. It was for a long time believed that the disease was caused by the combined attack of two fungi: *Pythium* and *Rhizoctonia*, but recent studies have shown that the disease is transmitted by an insect of the family Jassidae, *Recilia neivae*, that feeds primarily on grasses.

•Other leaf diseases of minor incidence are:

early leaf disease (or anthracnose), caused by a variety of fungi including *Glomerella cingulata*, *Botryodiplodia* spp. and *Melanconium elaeitis*; *Corticium*

Harvesting. A fresh fruit bunch (FFB) weighs on average 20-30kg depending on the age of the tree. Oil content of fruit pulp is 50-60% or 20-22% of the bunch weight. Oil content of kernels is 48-52% of the kernel weight, or 2-2.5% of the bunch weight. Under optimal conditions yields may reach 25-30 t FFB/ha/y, and with an average extraction rate of 21-23% this corresponds with an approximate yield of 6 tons of oil per hectare. Under sub-optimal conditions the average yield drops to 4-5 tons of oil per hectare. The first 3-4 years, the production of the young palms is often small, of poor quality, and sometimes even not economic to be harvested. Full production starts from the 6th year onwards; it reaches its maximum 4 to 6 years later, and remains high for another 10 years. It begins to decline when the trees are 20-22 years old. In industrial plantations trees are cut and replaced after approximately 25 to 30 years, both because at that moment the production decreases and the trees become so tall that harvesting is difficult. Fruit bunches are harvested by hand (machete), usually once every fortnight, and brought to the border of the road in a collection point, from where they are transported by truck to the oil mill.

Milling and Oil Processing. The purpose of milling the palm fruit or, more precisely the fruit + stalks, is to extract the oil from the pulp and separate the palm kernel from the rest. Palm oil from the fruits pulp is extracted mechanically by pressing. In contrast, palm kernel oil is extracted chemically by a solvent, like the subsequent refining operations

On industrial sites, the pre-treatment operations consist of taking delivery of the bunches at the factory and weighing them on the weighbridge, placing them on the unloading platform and loading the sterilization cages. The treatment itself consists of a series of operations. The goal of steam sterilization is to strip the fruits more easily, to disinfect them by killing any pathogenic microbes and rendering the lipolytic enzymes inactive so as to prevent the formation of free fatty acids. Stripping separates the fruits from the stalks. The next step is to mix and crush the fruit to produce a hot, homogeneous pulp. It is used to separate the pericarp from the palm nut and crushes mesocarp's oil producing cells fairly well. The temperature inside the mixer (a cylindrical vessel with a central, vertically mounted, rotary shaft), is kept at 95° C. The pulp is pressed in a screw press, a hydraulic press or a centrifugal press. The mixing and pressing

constitute the oil extraction proper. The crude oil is decanted in the settling tank. The aim of this is to separate the oil from the water, dissolved sugars, salts and cell compounds originating from the mesocarp; the pure oil is skimmed off the surface while any impurities sink to the bottom. The clarified oil is then dehydrated to lower its water content to 0.1-0.2% for better and thus stronger storage. After pressing, the oil palm seeds are stripped from their fibers and dried. The kernels are separated from the shells by crushing. This part of the factory is called the crushing unit. The shells and fibers are burned in steam boilers and the kernels are bagged for export and/or chemical extraction of the palm kernel oil. It is important to know that modern, industrial-scale oil mills (capacity 5-20 t/hr) are self-sufficient energy-wise, meeting their energy needs by burning the fibers and shells in their boilers. In smaller installations or factories with less efficient boiling, mixing and pressing technology, additional energy must be supplied by wood, charcoal, fuel or electricity.

Traditionally, palm oil is extracted by hand, except for the pressing, which is performed by a screw press or a hydraulic press in the artisanal process. The methods vary greatly, depending on the region, dietary habits and end use of the oil, but basically consist in quartering the bunches, followed (immediately or after 2 to 4 days) by detaching the fruits, heating them in water, crushing and pressing them (by hand, a hand press or a motorized press), collecting the crude oil and clarifying it by prolonged boiling followed by settling. After a second skimming, the clarified oil may be boiled again to expel as much water as possible. This long, laborious process yields a poor quality oil, according to prevailing international standards, but one that is often more valued by the local population than industrial oil. Similarly, the extraction rate is lower than in the industrial process, the oil content varying from 10 to 18% by bunch weight, depending on the type of press that is used. The only mechanized elements in this type of extraction - mainly the presses - are manufactured locally or in the region while the motors are imported. The energy source is generally wood.

4. CITRUS

Citrus spp. are natives of the subtropical and tropical regions of Asia and the Malay Archipelago. They have been cultivated since ancient times, and spread to other regions of the world, including the Mediterranean, South America and southern states of USA such as California and Florida, where suitable soil and climatical conditions exist. The citrus growing belt follows the equator, extending either side of it about 35 North and 350 South latitude. It embraces tropical, subtropical and the intermediate zones.

~~The most important~~ ^{major} commercial species of citrus fruits are:

- Sweet oranges (*Citrus sinensis*)
- Limes (*C. aurantifolia*)
- Grapefruits (*C. paradisi*)
- Lemons (*C. limon*)
- Mandarins (*C. reticulata*). These are often called tangerines.

Planting density and expected yield

Tree spacing is affected by factors such as the species of citrus concerned, the cultivar, type of rootstock, environment, size of the orchard and the management practices the grower will be using. For example, if a grower uses machinery, he must leave enough space between the rows for the machines to pass when the trees are mature. Site quality in terms of soil characteristics and water availability should be considered. Expected lifetime of the orchard is also important and may be influenced by freeze potential, disease incidence, or non-agricultural development potential. Thus, decisions must be based on a number of situations.

Spacing of 6 - 7.5 m between rows and a middle width of 2 to 2.5 m provides adequate access for production and harvesting operations. Within this range, more vigorous trees, such as: grapefruit, lemons, tangelos, and other varieties with more spreading growth habits should be planted at wider spacings than oranges.

Spacing wider than 7.5 m take longer to fill their allocated space, thus reducing early yield potential.

Spacings between rows as close as 4.5 m can be managed with conventional production equipment with timely row middle hedging, however, fruit handling at these closer row spacings becomes a problem.

Spacing in the row of 3 to 4.5 m is considered suitable for new plantings. Tree vigor, site selection and external fruit quality requirements again are important considerations within this

5. RUBBER

The natural rubber tree (*Hevea brasiliensis*) is a native of South America introduced to Southeast Asia during the 19th century. Rubber trees can grow to a height of 18 to 39 metres and they grow best in warm and moist climate ranging from 70-95 ~~Fahrenheit~~ ^{Fahrenheit of 21-35} Centigrade with an annual rainfall of 80-120 inches (2,000-3,000 mm).

The young rubber plants are raised in nurseries for 6 to 8 months and budded with bud scion from identified source before transplanting in using population of 400 to 555 trees per hectare. In five to seven years, the trees have stems of 18-20 inches (45-50cm) circumference and are ready for tapping or harvesting. Tapping can continue up to 40 years or beyond depending on the techniques of management.

Rubber is one of the top 5 priority commodities of the Department of Agriculture. Aside from generating employment in the rural areas and planting rubber in idle hilly lands and uplands will enhance environmental rehabilitation, being a good plant species in sequestration of carbon dioxide.

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Climatic and Soil Requirement

Since rubber cultivation is confined to the humid tropical zones, particularly South-East Asia, rubber is not grown on such a diverse range of soils as many other tropical crops. It is frequently grown on leached acidic inland soils of the humid tropics (latosols) which are often well structured and deep profiled but low in nutrients, or on coastal clay soils which are usually richer in nutrients (particularly those formed under a marine environment) but on which drainage problems are encountered. Rubber is also grown extensively on basaltic latosols which are deep, well-structured soils, higher in cationic nutrients than the latosols derived from other parent materials and with a higher reaction, but nevertheless still on the acidic side. Obviously nutrient requirements and fertilizer practice are given further on. The optimal pH for rubber lies in the range pH 4 to pH 6.5, but the crop tolerates pH in the range 3.8-8.0. Young seedlings tend to be more sensitive to low pH than mature trees. Soil pH above 8.0 definitely causes growth retardation, but most humid tropical soils are acidic in reaction.

Soils of the humid tropics which are less suitable for rubber, but are nevertheless often planted with the crop, are peat soils (in which tree support is a problem and various nutrient deficiencies occur), acid sulphate soils

Land and climatic requirement

1. flat/plain land well-drained ~~with deep water table~~
2. gently sloping/undulating to rolling terrains
3. soil pH of 4.0-6.5
4. good soil aeration
5. topsoil containing abundant organic matter
6. temperature range 20-34 Centigrade, average 27-28 °C
7. 80% atmospheric humidity with moderate wind speed
8. average rainfall of 2000mm evenly distributed throughout the year
9. bright sunshine amounting to about 2000 hrs, ~~with~~ at least 6hrs/day
10. no distinct dry and wet seasons

~~Land~~ Preparation

Adequate clearing of the planting site is recommended to avoid competition for nutrients with weeds and second growth forest trees. In flat cultivated areas, plow the area twice before laying out. In hilly lands, terracing is encouraged following the contour lines. Terracing must be done to prevent soil erosion. Plant spacing largely depend on the topography of the area and the possibility of planting intercrops

Plant propagation practices

Select seeds that are fresh, big, heavy and shiny. Before sowing the seeds, they must be soaked in water overnight. 10 kg of seeds can be germinated in every one-square meter of seedbed, 10cm thick, with fine river sand or sawdust as seedbed medium, and with partial shade. In sowing the seeds, they should be pressed firmly into the seedbed until the top of the seed is level with the surface, then cover the seeds thinly with the medium to prevent exposure to direct sunlight. Water the seeds twice a day and in 10-14 days, the seedlings can be pulled and planted into polybags or in ground nursery.

Planting

Budded seedlings ready for planting have matured top leaf whorls with healthy and vigorous appearance with dark green color. The size and shape of the planting hole would depend largely on the soil condition and planting materials. In fertile and light soils, holes should measure 25-30 cm in diameter and 40-45 cm deep. In poor and heavy soils, bigger holes are required - 40-45 cm diameter and 50-60 cm deep. Further, if the land is currently being cultivated, smaller holes will do, but in uncultivated land, bigger holes are needed.

Tree management (Pruning)

Pruning must be done regularly to develop a smooth trunk without branches or large scars on the stem/trunk along the optimum height of 2.5-3.0 meters from the ground. This should result to the bigger tapping panel. Pruning/cutting the top portion of the tree is not a recommended practice.

Nutrient Management

The first three years of plantation ~~establishment~~ **is** the most critical period where complete nutrition should be provided to the plant. Ideally, fertilizer application should be based on the results of soil and plant tissue analysis. To ensure that the optimum amount is applied. Folia, broadcast and ring methods of fertilizer application can be used and application is made at the start and before the end of the rainy season. Organic fertilizers can be used – with the rate of application based on the kind of organic fertilizer and recommendations.

Pest Management

Weeds in rubber farms can be controlled by line weeding, slashing, round/ring weeding, and the use of herbicides. Weeds in rubber areas must be controlled or minimized to prevent stunted growth of the rubber trees and to prevent fires during the dry season.

There are about six foliar fungal diseases of rubber: 1) bird's eyespot; 2) powdery mildew; 3) leafspot; 4) leaf blight; 5) anthracnose; 6) algal spot.

Harvest Management (Exploitation system)

Rubber trees are ready for tapping when the trunk circumference at 75 cm from the stock-scion union reaches 40-50 cm in circumference. It requires around five (5) year growing period under good management. Tapping should start on the 6th or 7th year after planting.

6. COCONUT

125 topic

Coconut palm (*Cocos nucifera* L.) is a member of the family Arecaceae (palm family) and the only species of the genus *Cocos*. Coconut fruit is a drupe not a true nut. A drupe is a fruit with a hard stony covering enclosing the seed like a peach or olive and comes from the word "drupe" meaning overripe olive. Such as other drupe fruits, it has three layers: exocarp, mesocarp and endocarp. The exocarp and mesocarp make up the husk of the coconut. The mesocarp or "shell" thus exposed is the hardest part of the coconut and is composed of fibers called coir. The shell has three germination pores (stom-) or eyes that are clearly visible on its outside surface once the husk is removed. The endocarp is the hard, woody layer that surrounds the seed. It is a large palm, a perennial crop classified as a fibrous one-seeded drupe, growing up to 30 meters (98 ft) tall, with pinnate leaves 4-6 meters (13-20 ft) long and pinnate 60-90 cm long; old leaves break away cleanly, leaving the trunk smooth.

Origin

Coconut palm is not indigenous to Nigerian agriculture out of the humid tropics; Although it is known to grow under diverse types of climate and is highly adaptable, are usually grown along the sea coast and in plain grounds, with the aid of its husky exocarp, they can float on the ocean for months and still germinate when beached, hence they may have arisen anywhere between the eastern Indian and western Pacific oceans. Prior to the age of discovery, coconuts were dispersed from East Africa to the Pacific coast of Panama.

~~Sub~~ Variety

Coconut palms have two natural subgroups simply referred to as;

i. Tall and

ii. Dwarf.

~~Most~~ Most commercial plantings use high yielding varieties. Many varieties of coconuts *C. nucifera* are being cultivated in many countries. These vary by the taste of the coconut water and color of the fruit, as well as other genetic factors, some of them includes:

i. Dwarf yellow coconut ii. Dwarf orange coconut

iii. Golden Malay coconut

iv. Dwarf green coconut

v. Fiji Dwarf (Niu Leka)

vi. Green Malay coconut

viii. NIGERIAN COCONUT

viii. Yellow Malay coconut.

Genetically Improved Varieties

There have been developments of varieties of coconut which yield 80-100 nuts per palm per year under proper management as against 30-45 nuts from the unimproved coconut. Below are some of the varieties prevalent in Nigeria and their yields per relative yields per year.

i. West African Tall: This variety produces 80-100 nuts/palm/year with an estimated copra yield of 3.3 metric tons per hectare per year.

ii. Dwarf Green: This flowers 3-4 years after yield establishment and produces 81 nuts/palm/year.

iii. Hybrid Coconut: This is a cross between the dwarf variety and the tall. The hybrid flowers 5 years after planting and produces 94 nuts/palm/year with an estimated copra yield of 3.5 tonnes/ha/year.

iv. Malayan Dwarf Yellow: This variety is used mainly for ornamental purposes. It is an introduction from Malaysia which flowers 5 years after planting and produces 84 nuts/palm/year with an estimated copra yield of 2.1 tonnes/ha/year.

v. Malayan Dwarf Red: This is also an introduction from Malaysia. It flowers about 3 years after field planting and produces 91 nuts/palm/year with an estimated copra yield of 2.3 tonnes/ha/year. This variety is also mainly used for ornamental purposes.

Economic Importance

Coconut is one of the most important and useful palms in the world, with Indonesia, Philippines and India respectively ranking as the Topmost producers of coconut in the world. It is fondly referred to as the "tree of life" for its important role in smallholders' livelihoods as a direct source of cash income, nutrition, and materials (Warner, 2007). In the Nigerian Agricultural industries, there are potentials to make major contributions to the economic and industrial development of the nation, especially with the wild range of industrial application of most of the products like Coconut. For a crop not indigenous to Nigeria, she is blessed with coconut trees which could be harnessed for industrial development through which the quality and standard of living of the people can be improved.

Coconut occupies a pre-eminent position in the Nigerian economy in providing employment to a large number of people living in the coconut belt if special consideration is given to its cultivation.

There will be a reduction in the unemployed population if the opportunities provided by the numerous application of the coconut palm are fully tapped.

(2) The planning, harvesting, and processing of the coconut will not only provide business opportunities to thousands of people in the region where it is cultivated, but it will also offer a wide range of investment opportunities that are economically attractive to the people at home and abroad.

(3) The role of the coconut in food production, foreign exchange earnings, raw materials for industries, income and employment generation to millions of Nigerians including women and young people make it a very crucial asset for National Economic Development.

(4) It is an important crop in the agrarian economy of many countries of the world providing food, drink, and shelter.

Uses

The uses of coconut vary and range from being used as food, industrially, decorative purposes, etc.

Some of its uses include;

- i. For the making of candies
- ii. For cooking (coconut rice)
- iii. For drinks. (coconut water)
- iv. As fuel wood (the mesocarp, Husk)
- v. For decorations
- v. For making coconut oil (Copra) etc

CULTIVATION

Land Preparation

The selected land should be completely cleared ensuring minimal disturbance of the topsoil layer. Compacted soils should be ploughed and stumps removed to provide adequate drainage. Land areas with rocks, clayey soils, and waterlogged areas should not be used. The field borders should be established along with fence lines, access roads, bridges, crossings, main drains, and channels, ponds or wells for irrigation and other infrastructure to ensure proper supply of moisture.

Soil Requirement

The ideal soil conditions for better growth and performance of the palm are loose well-drained soils about 50 - 100 cm deep with good water holding capacity. However, coconut can be grown under different soil types such as loamy, laterite, coarse, sandy, alluvial, clayey and reclaimed soils of marshy lowlands.

Planting Season

The coconut is a plant that is largely dependent on the presence and use of water, hence, the planting is done at the beginning of the rainy season. The propagation of coconut is mostly done using seedlings which are grown and nurtured in a Nursery. The nursery should be planned with respect to the seasonal changes. Seedlings should be 8 - 10 months of age before transplanting for better growth and development. They are planted immediately or at least 3 days after removal from the nursery to avoid mortality.

Spacing

For each planting hole, an area of soil 1 m in circumference is cleared of all vegetation and leveled. In planting the seedlings, the holes should be dug $0.3 \times 0.3 \times 0.3$ m for well fertile soils and in marginal soils, holes $0.9 \times 0.9 \times 0.9$ m are recommended. When planting, the planting holes can be dug at least 1 month before planting and filled with a mixture of topsoil, coconut husks, wood ash and well-decomposed manure which is then allowed to settle.

There are about three major planting space patterns which are;

- i. Square pattern,
- ii. Triangular pattern and
- iii. Single Hedge pattern.

Below are some of the spacing and the yield expectancy for planting in these patterns;

- i. For square plantings space, 8 m (158 seedlings/ha),
- ii. For triangular plantings space 8.5 m (158 seedlings/ha),
- iii. For rectangular plantings space 7.3×8.5 m (164 seedlings/ha)

Fertilizer and Nutrition

To ensure efficient growth, the seedling is placed in the hole and backfilled with topsoil mixed with 30g NPK (12-24-12). The seedlings are handled carefully so that the sprout is not damaged. The nut is covered with soil mix which is gently pressed into place so that the central bud and the collar of the shoot are above the soil level. The surrounding soil around the seedling is raked then

the seedlings are irrigated adequately. Irrigation is recommended immediately after planting. As the plant grows, the soil is backfilled to the surrounding soil level.

For Pest

The major pest that affects the cocoa plant includes:

1. Rhinoceros beetle (*Oryctes rhinoceros*)
2. Coconut eriophyid mite (*Aceria guerreronis* Keifer)
3. Red palm mite (*Raoiella indica* Hirst)
4. Red palm weevil (*Rhynchophorus ferrugineus* Olivier)
5. Leaf-eating or black-headed caterpillar (*Opisina areolaria*)
6. Scale insect (*Aspidiotus destructor* Signoret)
7. Coconut mealybug (*Nipaecoccus nipae* Maskell)

Note they are all causal agent to the diseases faced by the plants at respective times and seasons.

Diseases

Some major disease that affects the cocoa plant includes;

1. Red ring disease
2. Lethal yellowing
3. Bud rot (Heart rot)

Harvest

Cocunut is a Perennial crop that produces every single year from its first fruiting to points of maturity. However, it takes about six to ten years before cocunut can start fruiting and producing its nuts. Cocunut take about 12 months after pollination to fully ripen to the dry nut stage. Twelve-month-old nuts can be harvested for seed.

The mature nuts are harvested when at least one nut in the oldest bunch starts becoming dry. In Tall varieties, it takes 11 - 12 months from fruit set for a nut to mature whereas in Dwarfs, nuts will mature in 10 - 11 months after the emergence of the inflorescence. These nuts can be easily identified when the epicarp of one or two nuts in the bunch starts turning brown. Traditionally, you know the ripe nut by the sound they make when hit with a harvesting knife, it makes a dull sound when it's not ripe and a ringing sound when it is ripe.

Harvesting however is done at various points and times with respect to the used for the nut harvested. Harvest for cocunut water is done at about the 9-11 month stage. For cocunut oil, at the 12 - 13-month stage, the water (liquid endosperm) converts to the white meat (solid endosperm) lining the inside of the endocarp (shell). Nuts harvested at this stage are often dried to produce copra or wet processed to extract cocunut oil. In practice, the harvesting cycle varies from 45 - 60 or 90-day periods with high copra and oil recovery. Harvesting too early can affect the quality of the copra in terms of oil content.

There are two common methods of harvesting coconut: the pole and the climbing methods. Although both the pole and the climbing methods of harvesting require considerable experience and skill to be performed safely and efficiently, each has its own advantages and disadvantages.

Also, in many cases, dry nuts are allowed to free-fall and then collected from the ground beneath the trees prior to their processing.

Handling and packing and Processing Opportunities

In Nigeria today, there are several opportunities that are available to individuals in the value addition (processing and packaging) of coconut. With value addition, coconut has much potential from itself, as it can be processed into various forms with readily available and consuming markets in Nigeria.

Coconut can be processed into various products that can be used industrially or as food. Some of these products from coconut processing include:

- i. Coconut chips
- ii. Coconut candies
- iii. Crude coconut oil etc

Industrially, coconut extracts are processing into the making of

- i. Cosmetics (Body lotions and Creams)
- ii. Also, it is processing into the making of some fruit drinks in Nigeria (chivita, chi-exotic etc)

7 AVOCADO plants

Avocado fruits are also known as Avocado. Fruit is "*Persea Americana*" and it belongs to the family "Lauraceae" and genus "Persea". "Elegance Pierce", which is mostly grown in Mexico, Peru, India, Kenya, Persia, Chile, USA, Colombia, New Zealand, Dominican Republic, Indonesia, Brazil and many other countries, all over the world. Avocados are also known as the butter fruits is an evergreen tree of the Laurasi species.

Avocado fruit contains high fatty acid content, but it has very little cholesterol levels. Many people avoid eating avocado because of the high calcium contained in them because they feel that they can gain weight by consuming it. However, compared to butter, the amount of calories in it is very low and because of which it is known as butter fruits.

Climate

Frost-free areas are preferred, but mature trees will tolerate temperatures as low as -4°C for short periods without damage unless flower buds, flowers or fruit are present. Trees can also tolerate temperatures as high as 40°C for short periods, however fruit damage and losses can be expected. Growers in hot areas use overhead sprinklers (assuming good quality water is available) for evaporative cooling to preserve the crop during heat waves.

For avocado trees to produce flowers, a period of about 4 weeks of relatively cool temperatures needs to occur in autumn/winter. For fruit set to occur, periods of 3 days in a row are needed at flowering time during which the temperature does not drop below 10°C .

Soil Requirement

Avocado tree or plants are able to grow in all types of soil including reddish-brown soil to the red soil, dark brown soil along with subsoils. Avocado trees are very sensitive to the fertile soil which also has good drainage capacity. A well-drained soil is helpful in good aeration in the soil. However, growing avocado plants or trees on reddish-brown soil, dark brown soils are considered as the best suitable for optimum fruit production.

However, high yielding varieties of avocado trees or plants prefer medium and sandy soil having pH, ranging between 5.7 to 6.7 is considered as the best suitable soil for excellent fruit production. Also, avocado plants or trees also thrive their best in the soil which contains clay content, about 40 % of the soil. So, avoid the water logging problem in this commercial fruit farming.

Tree spacing

Tree spacing depends on whether you go for high, medium or low-density plantings.

- High-density plantings (more than about 500 trees/ha, e.g. 6m by 3m) give earlier returns, but cost more to set up. Because these plantings crowd together much sooner, you need to

start pruning trees earlier. In warmer areas high-density plantings will be more difficult to manage and may be impractical because tree growth is faster.

- Low-density plantings (200 trees or less/ha, e.g. 10m by 5m) provide lower early returns per unit area but are easier to manage. However at low-density, trees are generally allowed to grow larger which will:

- increase picking costs
- reduce picker safety
- produce less fruit and of lower quality.

As a compromise, you can choose medium-density spacings (e.g. 9m by 6m).

Planting times

In areas where frosts occur, we recommend planting during September–October. In warmer areas, April–May is best to avoid very hot weather whilst plants get established.

Fertilizing

Correct nutrition is critical but fertilizer treatments should be based on the results of regular soil and leaf analysis. In less fertile soils, soil analysis results may suggest adding lime, dolomite, phosphorus and selected trace elements before planting. If you need to apply fertiliser or manure before planting, do so 6 months before so it's broken down and taken up by the soil before planting. These fertilizers pose less risk to tree roots. You should never use nitrogen, potassium or animal manures on or just before planting. Fertilizing soon after planting can burn the tender feeder roots of the young tree.

You should pay particular attention to levels of:

- nitrogen
- boron
- calcium
- zinc
- soil pH.

Nitrogen encourages tree growth and canopy health. Boron is important to add before and during flowering for fruit set, calcium is important for fruit quality and zinc is an important trace element that is often deficient or unavailable in Queensland soils. Soil pH (measured in water) of about 5.5 is considered best but this level is impractical to reach in calcareous soils. Trees can grow successfully in pH of up to about 7.5, although iron chlorosis can be an issue.

Pruning

Without canopy management, yield and fruit quality will deteriorate and trees will become more physically difficult to spray and harvest. The type of canopy management practice will depend on tree density and the growth habit of the variety. The aim of canopy management is to:

- Keep trees to a practical size
- allow sunlight into the canopy to encourage flowering and fruitset
- rejuvenate the branches to keep them productive.

For a spreading, multi-branched variety such as Hass the most universally successful approach is to selectively remove 1 major limb per year. e.g. for a tree with 4 major limbs, no branch will ever be older than 4 years. Upright varieties, such as Lamb Hass trees, will need to be 'topped' every 2 or 3 years to keep height down, as well as some selective pruning of side branches.

Conduct major pruning after harvesting, i.e. during autumn or winter. Pruning too early in autumn (whilst temperatures are still high) could result in counterproductive rapid regrowth, whilst pruning too late (shortly before hot days are expected in spring or early summer) could result in sunburn to newly exposed limbs. We recommend seeking specialist advice and experiment with the timing and severity of pruning to find the most appropriate practice for your orchard.

Harvesting

Usually, this fruits crop takes around more than 10 yrs to become ready for harvesting. However, the commercial avocado varieties are able to give fruits in the 4th and 5th year of plantation. But, harvesting fruits at the proper time definitely yield higher fruit production.

So, avocados should be harvested at their mature age. To determine the maturity, observed the color change in the fruits. There are two varieties of avocado, the purple and green. Generally, the purple variety changes color from purple to maroon whereas the green used to change color from green to yellow. However, start harvesting when the seed coat starts changing from yellow-white to dark brown color. Also, try to soften these fruits after harvesting for about 10 days so that fruits ripe a well.

Yield of Avocado Farming

The yield of avocado farming or cultivation depends on numbers of key factors such as the variety, plant age, soil type, climate conditions along with the farm management skills. Practiced for the cultivation.

However, averagely, it is possible to obtain fruits around 15 to 20 tonnes per unit hectare after 6 to 7 yrs of the plantation on the main field.

Avocado Tree | Pests

The following is a list of some pests and insects, which commonly found in the avocado cultivation.

- Fruit-spotting bug
- Mealybugs
- Helopeltis

- Mites
- Queensland fruit fly
- Red-shouldered leaf beetle
- Red-banded thrips
- Swarming leaf beetles

Controlling Measures

Presence of any of these may lead to lower down the fruit production up to many times. So, these pests and insects should be controlled by a suitable way. Controlling these through the use of particular pesticide is considered as the best way of controlling them.

Avocado Tree | Diseases

The following is a list of some diseases, which commonly found in the avocado fruit cultivation.

- Root rot (Most common, caused by attack of Phytophthora cinnamoni)
- Bacterial soft rot
- Cercospora spot (attack on roots, leaves, and fruits)
- Black spot (attack on roots, leaves, and fruits)
- Anthracnose
- Phytophthora root rot
- Stem-end rot
- Pepper spot

Controlling Measures

There are numbers of fungicides and liquid fertilizers are available in the market, which has effective result in controlling all types of diseases. However, diseases in any crop occur due to over irrigation, lack of care in planting, and deficiency of any organic matter. So, these should be corrected at the proper time to avoid the loss.

Health benefits

However, ~~here~~ below is a list of some of the health benefits of eating avocado fruits. Let's learn the list:

1. Avocado fruit contains fibers; soluble and insoluble that is beneficial for maintaining a healthy digestive system.
2. Avocado fruits are also beneficial for the diabetic person.
3. Avocado fruits are high in fatty acid but are low in cholesterol level, so eating this fruit is also beneficial for the pregnant women, keeping healthy and happy.
4. Eating this fruits has an excellent result in weight loss and weight management.
5. Avocados are also good for keeping skin soft, fair and healthy.

6. These fruits are low in cholesterol level so these are helpful in lowering the cholesterol level and also reduce the danger of stroke.
7. These fruits are also helpful in relieving from the arthritis diseases.
8. These are the excellent source of anti-oxidants, hence are beneficial for the eye health.
9. Avocados are also helpful in preventing certain kinds of cancers.
10. Avocado fruits are also beneficial in providing prevention from chronic diseases and are also good for blood regulation.

8. African breadfruit

The African breadfruit (*Treculia africana* Desf.) is a member of the Moraceae, a family that includes the more popular Polynesian breadfruit, *Artocarpus alatus* Fosberg (also called *Artocarpus communis*). The family contains over 50 genera and over 1000 species of tropical trees and shrubs, many of which produce edible fruits and seeds. The members of this family are much-branched trees, up to 20 m high, inhabiting tropical wet coastal plains and well-drained soils. The trees bear large fleshy fruits which are covered with wart-like growths. The African breadfruit (*Treculia africana*), is filled with seeds, the plant being cultivated primarily for its delicious ellipsoidal seeds. Seed production from the African breadfruit is considerable, a mature tree producing up to 30 fruits annually, each fruit yielding 5-10 kg of seed after processing.

Origin

Breadfruit and breadnut are not native to the Caribbean. The seeded breadnut, which is considered as an ancestor of breadfruit, originated in Papua New Guinea in the Western Pacific region. Thousands of years ago, as early peoples moved eastwards across the Pacific with breadnut, the seedless breadfruit developed and became predominant in the eastern part of the South Pacific. Most of the breadfruit in the Caribbean came from this region.

Cultivation details

Tolerant of a wide range of climatic conditions, it will thrive in most tropical and subtropical regions at elevations up to 1,500 metres. It is found in areas where the mean annual rainfall is within the range 1,250 - 3,000mm, and the mean annual temperature ranges from 22 - 35°C. Prefers a rich, deep, moisture-retentive soil and a position in partial shade or full sun. It is not a very light demanding plant, and will grow in a wide variety of soils. Plants start to fruit when about 4 years old. Under favourable climatic conditions and in a good soil, 120 - 200 kilos of dried grains can be harvested in a year from each tree; this would translate to between 5 - 10 tonnes per hectare. A fairly fast-growing tree. There is a striking variation in the number of fruit heads produced by trees belonging to *T. africana* var. *africana* (with large fruit heads).

Propagation

Propagation is by seedlings, which are sown in pots, and direct spot sowing at site. Pre-treatment of seeds is not necessary. It is also propagated through budding, cuttings and shield grafting. Using adult scions, budded trees have produced fruits with viable seeds within 2-4 years. With juvenile cuttings, one can obtain 69% rooting. Where climatic conditions are favourable, fruit collection is possible throughout the year.

Botany and growth

The breadfruit and breadnut trees are evergreen, perennial with a lifespan of 50 or more years. Under favourable growing conditions, these trees grow continuously and can extend upwards quickly, therefore, they may reach to heights of 20 m or more in less than 20 years. During early growth, breadnut seedlings grow taller than young breadfruit trees. The tree consists of an upright trunk which is usually single, but in breadfruit, there may be two or more main trunks, on which many lateral branches arise in groups, each consisting of 2 to 7 branches, from which smaller branches also arise as the tree grows. The tree has a pyramidal shape because the lower main branches in the canopy are longer and, sometimes are held more horizontally than the upper branches which are usually more erect. Latex is present in all parts of the tree and the wood is quite brittle. At the ends of the branches are 7 to 15 large, glossy, green leaves, with bright yellow to light green mid-rib and main veins, and short hairs mainly on the lower surface. Breadfruit leaves are commonly about 30 cm long and 25 cm wide, and for most varieties in the Caribbean, the margins are not entire but divided into five or more lobes on either side.

Varieties

Based on detailed field observations, three varieties are distinguished within the subspecies: *T. a. var. africana* (extending from Senegal to Southern Sudan and south to Angola, central Mozambique and Sao Tomé and Principe islands), *T. a. var. inversa* (Anambra State, Edo and Delta States, more abundant in the eastern states of Nigeria), and *T. a. var. mollis* (isolated localities in Edo and Delta states of Nigeria, Cameroon, Congo, and Gabon).

Their taxonomic differences are based mainly on the size of the fruit head (inflorescence) and the hairiness of branchlets and leaves. A striking variation exists in the number of fruit heads produced by trees belonging to *T. a. var. africana* (with large fruit heads) and *T. a. var. inversa* (with small fruit heads). The former is clearly superior in the weight of seeds produced, while the latter produces more fruit and also produces twice as many branches.

Temperature

Breadfruit and breadnut are tropical crops that grow best at temperatures between 210 and 320 °C; growth is retarded at lower or higher temperatures.

Rainfall

The crop thrives best in locations where the annual rainfall receipt is 1500 mm to over 3000 mm and well-distributed throughout the year. Breadfruit has a higher water requirement because of its shallower root system and larger canopy. Breadnut can tolerate drier conditions better although breadfruit will grow where rainfall receipt is as low as 1000 mm annually but well distributed.

Soil conditions

Breadfruit and breadnut trees can grow on a wide range of soil types but they prefer sufficiently deep, well-drained and moisture retentive soils. Soils that are prone to waterlogging or seriously eroded and shallow should be avoided. The soils should have fairly high organic matter content, adequate nutrient status and a pH of 6 to 6.5.

LANDPREPARATION

Land preparation should be undertaken during the dry season. Tall weeds should be removed manually or mechanically, depending on the terrain. The trees will be planted in rows, therefore, it is best to orient rows in an east to west direction to allow for maximum sunlight availability on all sides of the trees during the course of the day.

• For flat to gently sloping sites

Drainage is the key consideration on flat sites and therefore, poorly drained sites should be avoided if possible. If there are no drainage or other soil physical conditions problems, further land preparation may be unnecessary and tillage should be minimal. On flat sites that experience occasional, short periods of water-logging, drainage must be improved. The land may be disc-ploughed, formed into cambered beds and box drains installed, 50 cm wide x 30 - 45 cm deep between the beds. These in-field drains should connect to a main drain at least 1 m deep for rapid water removal. Where hard pans are the cause of impeded drainage and long periods of waterlogging during the rainy season, deep ploughing is recommended to break up the pan before preparing cambered beds and drains.

• For moderately sloping sites

Soil conservation and water management are two key concerns peculiar to upland sites. Breadfruit and breadnut orchards should not be established on slopes greater than 25° (moderately sloping) where there is greater risk of soil erosion and reduced water conservation, and orchard management operations and harvesting are more difficult to conduct.

Spacing

The selected spacing and arrangement in the orchard must be guided mainly by the expected height and canopy width of the trees at maturity because this will affect light availability and air movement. Tree size is determined by cultivar and environmental conditions. Other important considerations are the cropping system, proposed use of machinery in the field and slope of the site.

For a pure stand orchard planted with the traditional breadfruit cultivars or breadnut at a site where the environmental conditions promote rapid growth, for example, high rainfall or high fertility, the mature trees should be maintained at a maximum tree height of 8 m and maximum

canopy width of 8 m. The recommended spacing is 10 m x 10 m which would give a plant population of 100 trees/ha (40 trees/acre). However, with narrower canopy width e.g. 6 m x 4 m closer spacing of 8 m x 8 m or 8 m x 10 m may be used to achieve a plant population of 156 trees/ha (59 trees/acre) or 125 trees/ha (47 trees/acre) respectively.

Diseases

• Fruit rot

Rots are the major fruit disease problem in breadfruit, but it is much less prevalent in breadnut. It has been attributed mainly to two fungal pathogens, *Glomerella cingulata* (the immature form is *Colletotrichum gloeosporioides*) and *Phytophthora palmivora*. *Glomerella cingulata* causes anthracnose which first appears as round, brown spots on the surface of immature fruits and may be spread by insects such as bees, and by wind.

• Tree Dieback or Decline

Tree decline occurs in some parts of the Caribbean. The first symptoms are excessive leaf yellowing and leaf fall, followed by the dieback of the younger branches. This dieback eventually extends to larger branches and the trunk and can kill the entire tree.

• Root rot

This disease, which also affects avocado, is caused by a fungus, *Rosellinia* sp. The disease spreads rapidly, and trees do not usually survive. Recommended control measures are isolation of affected trees with ditches, application of lime to the soil and use of copper fungicides.

• Abiotic factors

Sunburn gives affected fruits a scorched appearance that renders them unmarketable. Fruits that are exposed to strong sunlight, especially at the top of the tree, are more susceptible to this problem than those in the lower canopy where there is more shade. As the orchard matures and more self-shading occurs, the incidence of sunburn should be lower.

YIELD

Breadfruit trees begin bearing in less than three years after field establishment, especially 'Yellow', while breadnut may take three years where the environmental conditions and management of the crop are very good. Trees come into bearing later under less favourable conditions. Both 'Yellow' and 'White' cultivars tend to have two production seasons, with harvesting extending from June to September, the major season, and from January to March, the minor season. The actual commencement of the season and its duration depends on several factors including the prevailing weather conditions, and crop factors e.g. cultivar. The first fruits to mature come at the top of the canopy where flowering and fruiting begins first because of

better light receipt than in the lower canopy. An 8 ft. maturity tree may produce more than 100 to 150 fruit depending on cultivar, age and management. Early fruit fall, approximately 6 to 8 weeks after bearing begins, may be as high as 25%. The other fruits reach maturity in 14 to 16 weeks by which time the majority attain a weight of 1.5 to 2 kg.

Harvesting

Breadfruit and breadnut fruits should be harvested at the stage of maturity that is required for the method of preparation for consumption. This requires some knowledge of the maturity indices for different stages of fruit development. Fruit size, skin colour, skin texture and the absence or presence of latex on the skin are the main indicators of the stage of fruit maturity for these species.

MATURITY INDICES FOR BREADFRUIT

Breadfruit is harvested at the following stages of fruit maturity:

Immature

Fruits are usually less than 12 weeks old, bright green in colour, very firm and still fairly undersized. The cut fruit exudes copious quantities of latex and the flesh, which is initially almost white, quick discolours. The flesh also has higher water content and the taste is very bland compared with more mature fruit. A small percentage of fruits is harvested at this stage for use mainly as an ingredient in soups e.g. 'Mannish Water' or for pickling as in 'Sousee'.

Green mature

At this stage the fruit has completed its development (typically by 14 weeks) and has reached the dimensions of a mature fruit but has not begun to ripen and is still very firm. The peel may be greenish brown with some latex stains, especially in 'Yellow' which has a smoother peel than 'White'. The cut flesh shows little or no discolouration and less latex flow than in immature fruit. The flesh is very starchy and not sweet to the taste. A large percentage of fruit is harvested at this stage in countries where the fruits are consumed mainly by boiling or steaming. Fruits that are to be exported are also harvested at this stage to prevent premature ripening.

Fully mature

The fruit at this stage may be 15-16 weeks old. The skin of 'Yellow' fruit changes to brownish yellow-green, the peel becomes very smooth and latex staining may be prolific (Photo 11 a), while the skin of 'White' fruit becomes light green and the peel is smoother with more latex stain than in less mature fruit of that cultivar, but it is less stained than 'Yellow' fruits (Photo 11 b). The cut flesh is light yellow and white to cream in 'Yellow' and 'White', respectively, and shows no discolouration or latex exudation. The flesh is firm and may be slightly sweet on